

ARK-455 — CORRECTION / ERRATUM (v1.1)

Issued: 2026-07-17 (UTC)

Applies to: ARK-455 v1.0 (Zenodo DOI 10.5281/zenodo.21408545; concept 10.5281/zenodo.21398675)

Nature of correction: Root-cause reclassification. **No raw data changed. No verdict rescinded.**

Summary

ARK-455 v1.0 recorded a **FAIL** verdict: in Arm 3 (timestamp), the “tampered” records were accepted by both independent verifiers at a 0% rejection rate, while all six other tamper targets were rejected at 100%. The v1.0 record speculated the cause was **either** “a generator flaw (tampering strategy insufficient to break the signature)” **or** “a verification logic gap.”

A post-publication source audit has now **resolved that ambiguity definitively**. The cause is the **first branch: a generator (test-harness) defect**. The Arm-3 timestamp “tamper” was a **no-op** — it never altered the record. It is **not** a verifier defect and **not** a specification gap in the signed payload.

The FAIL verdict stands as an honest, recorded outcome. What changes is only the interpretation of the failing arm: it reflects a defective tamper generator, not a detection gap in the dual-verifier architecture.

Evidence (reproducible from the locked v1.0 code)

The Arm-3 timestamp mutation in `generator/record_generator.py` is:

```
elif field == "timestamp":
    # Parse, add 1 second, re-serialize <-- comment says +1 second
    dt = datetime.fromisoformat(tampered["timestamp"].replace('Z', '+00:00'))
    dt = dt.replace(microsecond=(dt.microsecond + 1000000) % 1000000) # <-- no-op
    tampered["timestamp"] = dt.isoformat().replace('+00:00', 'Z')
```

For **every** integer microsecond value `m` in `[0, 1_000_000)`:

```
(m + 1_000_000) % 1_000_000 == m
```

So `dt.replace(microsecond=...)` returns the **same** microsecond, the seconds field is never touched, and the resulting timestamp is **byte-identical** to the original. The comment (“add 1 second”) does not match the code (adds zero).

Consequently, Arm-3 “tampered” records are identical to Arm-1 (untampered) control records. Both verifiers therefore **correctly ACCEPTED** them — the signatures are

genuinely valid over unaltered records. The 0% rejection rate is the correct verifier response to a non-tampered input.

What the audit confirmed about the verifiers and the spec

- The `timestamp` field **is** included in the RFC 8785 (JCS) canonical form that is Ed25519-signed by the generator (`sign_record` signs the full 7-field record).
- Both verifiers (V1 JS, V2 Python) **do** reconstruct and canonicalize the `timestamp` field before verifying. A genuine post-signing change to `timestamp` **would** break the signature and be rejected — exactly as the other six arms show.
- Therefore the earlier external hypotheses that “timestamp was not in the signed payload” or “the mutation happened before signing” are **both incorrect** for ARK-455 v1.0.

Corpus scoreboard impact

- ARK-455 remains a **recorded FAIL** (the verdict is not rescinded; preregistration discipline preserved — nothing loosened after the fact).
- The FAIL is now correctly attributed to a **test-harness no-op**, not to a verifier or specification weakness. The dual-verifier architecture behaved correctly on all arms, including Arm 3.
- Follow-up **ARK-455b** (see the `ark-455b/` directory) re-runs the study with:
 1. a **corrected** timestamp post-signing mutation (a real change that breaks the signature),
 2. a **new pre-signing expired-timestamp arm** that tests validity-window/expiry semantics beyond bare signature checking, and
 3. a **mutation-effectiveness gate** that aborts the run unless every REJECT arm’s tamper provably alters the record — so a no-op can never again masquerade as a detection failure.

ARK-455b outcome (2026-07-17): PASS. With the corrected mutation, Arm 3 (timestamp) is now detected at **100%** by both verifiers, empirically confirming this erratum’s root-cause finding. The new pre-signing expired-timestamp arm is also rejected at 100% via the validity-window gate, and V1-V2 concordance is 100% across all 900 records. See `../ark-455b/RESULTS.md`.

Discipline note

ARK-455 v1.0 was published to Zenodo (DOI-live) under a working instruction to keep the record as a draft pending review. The record is honest and the raw data are sound, so no retraction is warranted — this erratum is the appropriate correction mechanism. The publication-gate instruction has been tightened for future runs (preregistered “draft, do not publish without human review” is now an explicit gate).